

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

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1.0 Abstract

The project aims to create a mobile application that helps parents manage and care for their children from the age of one day up to 7 years. The program provides solutions including: tracking growth and development, vaccination schedules, daily tasks with a gamification system, educational and entertainment content designed specifically for each age group, in addition to health and behavioral support through specialists. The project gathers everything parents need in one place, to provide a comprehensive and organized care experience.

1.1 Problem Statement

Parents today face a great challenge which is the absence of a reliable and integrated source that provides them with the necessary support for their children's care from the age of one day up to 7 years. They suffer from scattered information among multiple sources, and difficulty in following important timelines such as vaccination schedules and daily tasks. Also, they feel anxiety and exhaustion when dealing with common problems like sleep and behavioral disorders, or when facing simple health emergencies, because of a lack of specialized knowledge and the absence of a supportive community. This situation creates a psychological and mental burden on parents, and affects the quality of care provided to the child, which confirms the urgent need for a comprehensive and organized solution that gathers all these needs in one place.

1.2 Value Proposition

God willing, enabling new and experienced parents through a comprehensive digital platform that provides them with all the tools and knowledge they need to manage their children's care and development with confidence and ease. We combine specialized health and growth tracking, innovative educational activities, and behavioral support, in one place, which eliminates the need to search in multiple sources and saves time and effort, and makes the parenting journey more enjoyable and organized.

2.0 Introduction

In today's fast-paced world, the role of parents is one of the most difficult and most rewarding roles. Every day represents a new challenge and a new opportunity to watch our children grow. But with the multitude of tasks, and multiple sources, many parents may feel lost and dissatisfied with caring for their children. God willing, from here, we offer a comprehensive digital solution that was designed specifically to accompany parents on the parenting journey from A to Z. Instead of searching in dozens of sources, God willing, our application gathers everything they need in one place. Whether searching for a way to track a baby's growth, or an effective plan to improve the behavior of an older child, or even educational and entertainment content that suits their ages, the application is your reliable friend that provides you with the necessary tools, knowledge, and support.

2.1 Vision

To be the trusted partner for every Arab family in raising a conscious, healthy, and future-ready generation, by providing innovative and continuous solutions that support the parenting journey.

2.2 Mission

To provide a comprehensive mobile application that equips parents with reliable tools and content from experts, to enable them to follow their children's health, development, and behaviors with ease and confidence. We are committed to simplifying the parenting journey through smart and practical solutions for every age stage.

2.3 Main Objectives

- **Supporting Overall Health and Growth**

To provide reliable tools for tracking a child's growth stages, organizing vaccination schedules, and offering reliable health guidelines for parents.

- **Empowering Parents with Necessary Tools**

To provide practical solutions for managing daily tasks and facilitating the building of a healthy routine for sleep and nutrition, which helps save time and effort.

- **Enhancing Educational and Behavioral Development**

To provide educational and entertainment content designed specifically for each age group, in addition to behavioral and psychological support to help parents deal with common challenges.

- **Building a Supportive Community**

To create a secure communication platform that enables parents to exchange experiences, communicate with specialists, and get the necessary support.

3.0 User Research & Problem Analysis

To ensure the building of a digital application that meets the real needs of parents and solves their actual problems, it was necessary to conduct comprehensive research to directly understand the challenges they face. The data collected from the targeted users is the cornerstone for designing an effective and innovative user experience.

3.1 User Research

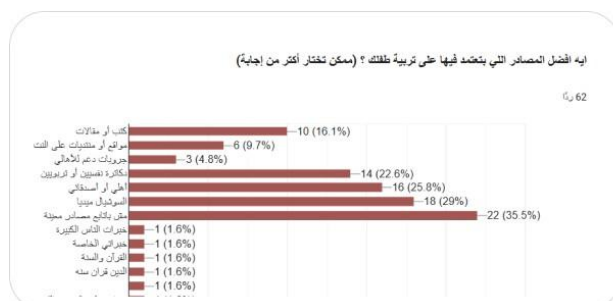
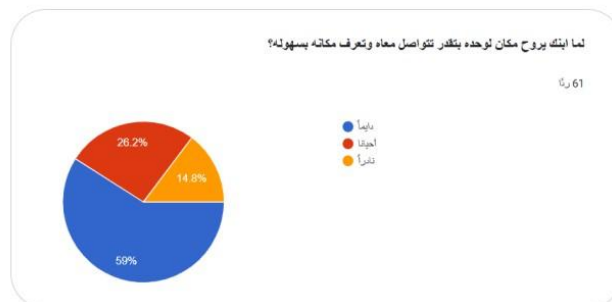
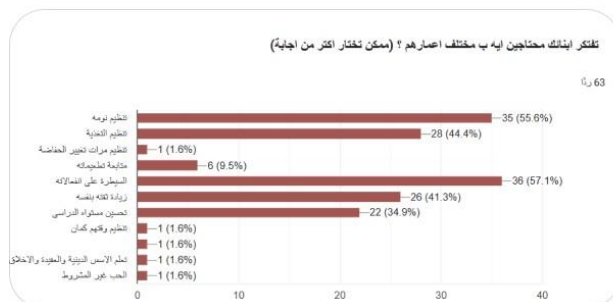
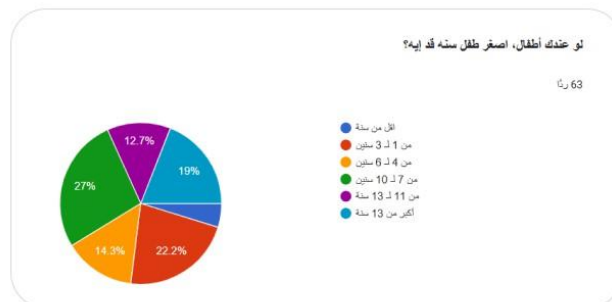
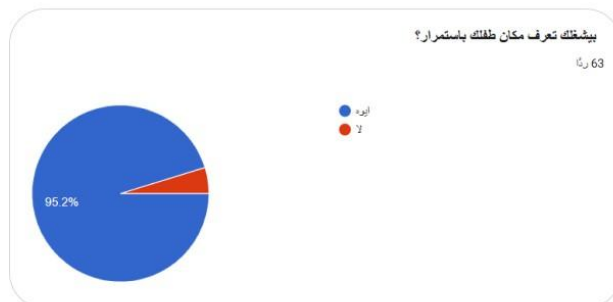
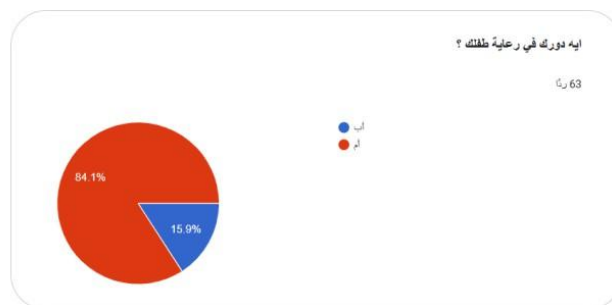
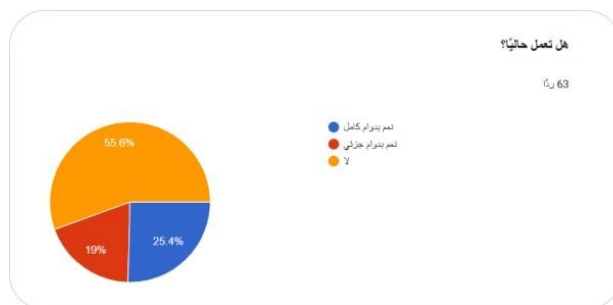
- **Surveys**

A survey was designed to collect quantitative data from a wide segment of parents about the most important problems they face in their children's care, and to determine their priorities in terms of the required features and functions in a specialized program.

- **Survey Goal:** The survey aims to collect quantitative data from a wide segment of fathers and mothers, to determine and classify the most important challenges and problems they face in their children's care. The survey also seeks to understand their priorities in terms of the required features and functions in a comprehensive digital program, which, God willing, ensures that our proposed solutions are based on real and urgent market needs.
- **Survey Methodology:**
 - **Target Audience:** Fathers and mothers of children aged between one day and 7 years.
 - **Survey Tool:** Google Forms was used.
 - **Number of Participants:** 61 fathers and mothers participated.
- **Survey Questions:**
 1. What is your role in your child's care?
 - Father
 - Mother
 2. Do you currently work?
 - Yes, full-time
 - Yes, part-time
 - No
 3. If you have children, how old is the youngest child?
 - Less than 1 year
 - From 1 to 3 years
 - From 4 to 6 years
 - From 7 to 10 years
 - From 11 to 13 years
 - Older than 13 years
 4. What things do you find difficult in your child's care? (You can write in detail)
 - [Open-ended answer box]
 5. What are the best sources you rely on for your child's upbringing? (You can choose more than one answer)
 - Books or articles
 - Websites or forums on the internet
 - Parent support groups
 - Psychologists or educational specialists
 - My family or friends
 - Social media
 - I don't follow specific sources
 - Other: [Open-ended answer box]
 6. Does it concern you to know your child's location constantly?
 - Yes
 - No
 7. When your son goes somewhere alone, can you communicate with him and know his location easily?

- Always
 - Sometimes
 - Rarely
8. What situation have you faced before in caring for one of your children and you didn't know how to act? And how would you act if it happened again?
- [Open-ended answer box]
9. What do you think your children need at their different ages? (You can choose more than one answer)
- Organizing their sleep
 - Organizing nutrition
 - Organizing diaper change times
 - Following their vaccinations
 - Controlling their emotions
 - Increasing their self-confidence
 - Improving their academic level
 - Increasing their self-confidence
 - Other: [Open-ended answer box]
10. How easy is it to find reliable and useful information about raising and developing children?
- Easy
 - Medium
 - Difficult
11. Do you have enough support from your partner/family/friends in raising your child?
- I have very enough support
 - I have enough support to some extent
 - The support is not enough
 - There is no support at all
12. How easy is it to find reliable and useful information about raising and developing children?
- Easy
 - Medium
 - Difficult
13. If you had a mobile application dedicated to your child's care, what would you like it to have?
- [Open-ended answer box]
14. We are waiting for your evaluation of the survey so we can improve it (highest rating is 5)
- 1
 - 2
 - 3
 - 4
 - 5
15. What are your suggestions so we can improve the survey?
- [Open-ended answer box]

Analysis of results



Interviews

In-depth interviews were conducted with a select group of fathers and mothers to gain a deeper qualitative understanding of their daily problems, their experiences in applying educational and health solutions, and their expectations from an application like our project.

- **Interviews Methodology:**

- **Target Audience:** Between 5 and 9 fathers and mothers were selected representing different age groups and who have children of different age groups.
- **Interview Duration:** Each interview duration was between 15 and 30 minutes.
- **Type Used:** Semi-structured Interview, based on a guide of questions or main points, but gives the interviewer flexibility to ask additional questions or rearrange them according to the flow of the conversation.

- **Discussion Questions (Open-ended questions):**

1. How many children do you have? And their ages?
2. What are the most difficult things you find in caring for your children?
3. How do you deal with your son's vaccinations? What problems do you face with vaccinations in general?
4. What are the problems you see in your children's sleep routines and what are their causes?
5. Can you describe your son's current dietary system for us? And what methods have you tried to improve it?
6. What are the difficulties you face in organizing your time between your son and the rest of the daily tasks? And why do you think it is difficult for you? Have you ever felt like you didn't know how to act in a certain situation with your child?
7. What is the thing that worries you most in their daily lives?
8. What is your experience in teaching your child basic skills such as using the toilet, dressing, and personal hygiene?
9. What is the most difficult behavior your child faces regularly (for example: severe tantrums, stubbornness, disobedience, sleep problems, or toilet training challenges)? How do you deal with this behavior now? And what methods have you tried that did not succeed? What is the feeling you have when you face this behavior?
10. What are the sources that help you find information about raising your child? Do you think the information is enough? Are there problems with the mentioned sources? What information do you try to search for regularly and might not find easily?
11. If there was a mobile application to help you organize and care for your child, what features would you like it to have? What is the most behavior you are currently trying to teach your child, and if there was an application, what is its role or how can it help you?

3.2 Problem Analysis

By combining the results of the surveys and interviews, we were able to build a clear and comprehensive picture of the challenges parents face, which forms the solid basis for designing our proposed solutions.

- **Vaccination Section Problems:**

- Following vaccination appointments and sometimes forgetting them.
- Parents' anxiety about the side effects of vaccination, such as high fever, and not knowing how to deal with them.
- The challenge of managing vaccination appointments for more than one child at different times.

- Difficulty finding places for vaccination when traveling.
 - **Sleep Section Problems:**
 - Parents suffer from frequent interruptions in their children's sleep, as the child wakes up crying or suddenly at night, which leads to parental exhaustion and disruption of the family routine.
 - **Daily Tasks and Time Management Section Problems:**
 - Difficulty finding daily time to play with children and strengthen the relationship with them.
 - The challenge of managing the tasks and schedules of 3 children of different ages.
 - Difficulty remembering the required daily tasks for the children.
 - The father's absence at the beginning due to lack of awareness of the importance of his participation.
 - **Nutrition and Child Growth Section Problems:**
 - Not knowing the child's increasing nutritional needs with age, which causes the baby to cry from hunger.
 - The child's refusal to eat in general, or eating very small amounts, and not sticking to a meal schedule.
 - The child's refusal to eat certain types of food, which affects the diversity of their nutrition.
 - **Behaviors and Upbringing Section Problems:**
 - The special challenges of dealing with a child with mild autism, with the goal of developing their skills and preparing them for school.
 - Stubbornness and aggressiveness among children at home, especially among young children.
 - Parents' inability to communicate with the child and understand the reason for their negative behavior, such as their refusal to go to school.
 - Parents' feeling of helplessness and inability to deal with the child's severe tantrums.
 - Difficulty answering the child's questions in a simple and convincing way.
 - **Educational and Entertainment Section Problems:**
 - The absence of suitable activities that occupy the child's time and develop their skills without relying on screens excessively.
 - Difficulty finding easy and suitable educational tools for the child to learn by themselves.
 - The challenge of reconciling different curricula for several children.
 - **Health Care Section Problems:**
 - Not knowing how to act quickly and effectively in emergency situations such as the child falling or getting injured.
 - Difficulty identifying foods that may cause allergies to the child when starting to try food.
 - Inability to make a decision about the child's health condition (for example, when their temperature is high) and whether it requires a visit to the doctor or not.
-

User Research Conclusion:

The results of the user research show that the challenges parents face are not uniform, but vary greatly according to the child's age stage. Each age group has its unique needs and challenges. Therefore, the problems that were identified have been divided into three main categories to ensure a deeper and more accurate understanding of each stage, which are:

- **Age Group (from one day - 1 year):** Its challenges are concentrated on the basic health and nutritional aspects.
 - **Age Group (from 1 - 3 years):** In which the problems of behavior and basic skills increase.
 - **Age Group (from 4 - 7 years):** In which the problems evolve to include complex educational and behavioral aspects.
-

4.0 Literature Review & Competitive Analysis

To ensure that our project is built on a strong and well-studied foundation, this section aims to provide a comprehensive overview of the current context of the child care field. This will be done through two main parts: first, a review of scientific literary research and theories related to child growth and development, and second, an analysis of the existing solutions and platforms in the market.

4.1 "Arabic Children's Applications on (Android): An Analytical Study of Usage Trends: Abdullah Mahmoud Adawi, International American University "

4.1.1 Abstract:

The study aims to analyze Arabic applications dedicated to children on the "Android" system, and to study their impact on enhancing the cultural identity and knowledge of the Arab child. With the spread of digital technologies, Arab children have begun to interact with various applications, some Arabic and others foreign. The research used a descriptive approach and content analysis to evaluate the educational, religious, and entertainment applications available on Google Play. The results showed that Arab children are more attracted to educational applications (such as teaching Arabic letters) and games, while their participation in cultural and religious applications is less. The study also revealed the dominance of foreign applications, which poses a risk to the Arab identity because children are exposed to content that is not compatible with their values. (Adawi, 2022, pp. 104-117) [1]

4.1.2 Problem

Weak Arabic Content: The scarcity of high-quality Arabic applications compared to their foreign counterparts.

Foreign Cultural Influence: Children's preference for foreign applications exposes them to different cultures and values that may weaken their connection to their Arab and Islamic identity.

Lack of Interaction in Educational Applications: Most Arabic applications lack visual appeal and interactive technologies that attract children.

Absence of Institutional Support: The lack of sufficient efforts from governments and educational institutions to support the production of distinguished Arabic applications. (Adawi, 2022, p. 112) , (Gerbner and Gross, 1976, p. 175) , (Mohammed Bin Rashid School of Government, 2014) [1]

4.1.3 Solution

Improving the Quality of Arabic Applications:

- Developing interactive applications that combine education and entertainment using animation and games.
- Enhancing religious and cultural content in an attractive way that suits a child's mentality.

2. Cooperation of Educational Institutions and Parents:

- Guiding schools and parents to use useful Arabic applications.
- Raising parents' awareness of the importance of monitoring the digital content their children are exposed to.

3. Increasing Competition with Foreign Applications:

- Arab companies investing in the development of high-quality educational and entertainment applications.
- Enhancing the presence of Arabic applications on platforms like Google Play by improving their promotion.

4. Enhancing Arab Identity in Applications:

- Producing applications that introduce children to Arab and Islamic history in an interesting way.
- Integrating heritage stories into applications to enhance cultural belonging. (Adawi, 2022, p. 115) , (Shrum, 2017, p. 8) , (Ball-Rokeach and DeFleur, 1976, p. 10)

4.1.4 Conclusion:

- Children who use Arabic applications for one hour a day showed:
- A 40% improvement in the Arabic language
- A 35% increase in cultural knowledge
- But 68% of parents do not guide their children to Arabic applications.

1. Main Results in Application Distribution for the Original Research:

- 28.5% Games
- 21.4% Educational
- 17.8% Religious
- 14.2% Cartoons
- 10.7% Stories
- 7.1% Cultural

2. Rates

- Educational Applications: 1.2 million downloads on average
- Games: 750,000 downloads
- Cultural: less than 50,000 downloads

3. The Gap with Foreign Applications:

- The foreign game Subway Surfers achieved 19 million Arab downloads
- While the most famous Arabic game achieved only 1.3 million.

4.2 Quality of Mobile Apps for Child Development Support: Search in App Stores and Content Analysis"

4.2.1 Abstract

The study explores the quality of mobile applications designed to support parents in promoting child development and health. With the rise of smartphone use, parents increasingly turn to mobile apps for parenting guidance; however, the credibility and effectiveness of such apps vary widely. Through a content analysis of apps from Google Play and Apple App Store, the research assessed features, cultural competence, user burden, and developer credentials. Findings revealed that most apps provide generic advice rather than personalized, evidence-based support. The authors highlight the need for evaluation frameworks to ensure that child development apps are reliable, culturally sensitive, and supportive of diverse families [2].



4.2.2 Objective

The main objective of this study was to examine the current landscape and quality of mobile applications designed to support child health and development. Specifically, the research sought to identify what types of apps are available to parents, evaluate the extent to which these apps promote developmental health, and analyze their design choices in terms of functionality, usability, and accessibility. In addition, the study aimed to uncover potential user burdens, such as financial costs, privacy concerns, and emotional stress, that might discourage parents from sustained use. Another important objective was to assess the cultural competence of these apps, including their ability to provide content that is inclusive, tailored, and sensitive to diverse family backgrounds. Ultimately, the study sought to generate insights that could guide future developers and health professionals in designing evidence-based, culturally responsive, and user-friendly apps that effectively support parenting practices [2].

4.2.3 Methods

The authors adopted a **content analysis approach** to evaluate mobile apps for early childhood health promotion. In January 2022, systematic searches were conducted across the Google Play and Apple App Store using keywords such as “baby health” and “child development app.” The inclusion criteria were: free to download, in English, and designed for children aged 0–5 years, covering tracking of health behaviors or milestones. A total of **149 apps** met the criteria and were coded based on features, content, privacy policies, developer credentials, user burden, and cultural responsiveness. Coding was guided by the **Bright Futures Guidelines** for pediatric health and the **User Burden Scale** [2].

4.2.4 Result

The initial search identified **1348 apps**, but after removing duplicates and irrelevant entries, **149 apps** were included for analysis. Of these, 36.2% offered feedback-based features (e.g., milestone progress reports), while 63.8% only provided static or generic information. Only 8.7% of apps referenced evidence-based guidelines, and just 9.6% included subject-matter experts in development teams. Cultural competence was lacking: only 20.1% offered multiple languages, and just 12.1% included diverse imagery. User burden was also significant, as many apps required in-app purchases or subscriptions, raising financial accessibility issues. Privacy was another concern, with 26.9% of apps not providing any data-use information. Overall, the study concludes that while apps hold potential for child health promotion, most lack personalization, cultural sensitivity, and evidence-based grounding [2].

4.3 “The effectiveness of early childhood development programs”

4.3.1 Abstract

This paper presents a systematic review of early childhood development (ECD) programs such as Head Start, focusing on their effectiveness in improving cognitive, social, health, and family outcomes for disadvantaged children aged 3–5. The review found strong evidence that such programs enhance school readiness, academic achievement, and IQ, while reducing grade retention and special education placement. Evidence for long-term social, health, and family outcomes was promising but insufficient. The study concludes that ECD programs are valuable community interventions that can improve children’s developmental trajectories and promote equity. [3]

4.3.2 Problem

Many children, particularly those from low-income families, begin school without adequate readiness skills. This lack of preparedness leads to poor academic performance, increased grade retention, placement in special education, and long-term risks such as school dropout, unemployment, and health problems. Existing research lacked sufficient evidence to comprehensively evaluate the broader social and health benefits of ECD programs. [3]

4.3.3 Solution

- Expand Access to Programs: Increase availability of publicly funded early childhood development (ECD) programs to reach more disadvantaged children.
- Integrate Services: Combine educational, nutritional, health, and family support within ECD programs to address multiple aspects of child well-being.
- Promote Multidisciplinary Collaboration: Encourage cooperation between health, education, housing, and social services to strengthen program impact.
- structure, program intensity, and parental involvement to maximize effectiveness.
- utilize the free version of the app.
- Promote Multidisciplinary Collaboration: Encourage cooperation between health, education, housing, and social services to strengthen program impact
- Support long-term and high-quality studies to better understand social, health, and family outcomes of ECD programs. [3]

4.3.4 Result

The review found strong evidence that ECD programs significantly improve **cognitive outcomes** (school readiness, IQ, achievement) and reduce **negative educational outcomes** (grade repetition, special education placement). Some studies showed long-term benefits in **social outcomes** (higher graduation rates, lower delinquency) and **family outcomes** (employment, income), though the evidence was limited. Overall, the study concludes that ECD programs are effective in promoting early learning and preventing developmental delays, with potential long-term societal and health benefits. [3]

4.4 “wide scale well-baby clinic mobile application”



4.4.1 Abstract

This study introduces *Sehhat Tefly*, a novel mobile health (mHealth) application developed in Egypt to support caregivers in monitoring the health and wellbeing of children under five years old. The app integrates seven key services: physical growth, developmental milestones, immunizations, nutrition, teething, safety and emergency measures, and reporting. It was developed through four phases: user requirements, application development, validation using the Mobile App Rating Scale (MARS), and implementation. The pilot study showed that the app was feasible, accessible, and well-received, offering caregivers a free, interactive, and evidence-based tool to support child health monitoring. [4]

4.4.2 Problem

Traditional well-baby clinics in Egypt are underutilized, with most visits focused only on immunizations, while growth monitoring, nutrition, developmental milestones, and early childhood health services remain neglected. Caregivers face barriers such as long waiting times, overcrowding, and limited comprehensive services, which hinder proper child health monitoring. Existing Arabic mobile apps were found to be limited, non-interactive, ad-driven, and not developed by recognized health organizations. [4]

4.4.3 Solution

To address the limitations of traditional well-baby clinics and the lack of comprehensive childcare applications, the researchers developed *Sehhat Tefly*, a novel mobile health application designed to cover all major components of well-baby services. The app integrates seven essential features: growth monitoring using WHO charts, developmental milestones assessment based on CDC tools, immunization tracking with reminders, nutritional guidance, teething monitoring, safety and emergency measures, and generation of health reports. It was designed with interactive features to provide tailored feedback and automated advice to caregivers, ensuring practical support in daily child health monitoring. The application was validated by pediatric and public health experts using the Mobile App Rating Scale (MARS) to confirm its quality, functionality, and usability. Finally, it was deployed on the Android platform in Arabic, ensuring accessibility, data privacy, and evidence-based information for caregivers across different sociodemographic backgrounds. [4]

4.4.4 Result

The pilot implementation of *Sehhat Tefly* demonstrated promising outcomes in supporting child health monitoring. Within four months, the app was downloaded 1,445 times, with 943 caregivers registering their children, indicating strong acceptance and demand. Expert evaluation using the Mobile App Rating Scale (MARS) produced an overall quality score of 3.7 out of 5, reflecting good performance in terms of functionality, aesthetics, engagement, and information quality. Caregivers highlighted nutrition, developmental milestones, and growth monitoring as the most needed services, aligning with the app's core functions. Overall, the study confirmed that *Sehhat Tefly* provides a feasible, accessible, and low-cost digital tool that can enhance caregivers' ability to track their children's health, detect potential issues early, and complement existing well-baby clinic services[4]

4.5 Privacy in Popular Children's Mobile Applications: A Network Traffic Analysis"

4.5.1 Abstract

Children increasingly engage with mobile applications available on major marketplaces such as Google Play Store and Apple's App Store. Although these platforms claim to provide child-friendly applications, concerns remain regarding the transparency and legality of personal data collection. This study conducted an in-depth technical network traffic analysis of 15 popular children's mobile applications, categorized across three age groups (0–5, 6–8, and 9–12 years). Using intermediary monitoring tools, we recorded and analysed outbound data traffic to identify potential privacy violations and assess the accuracy of disclosed data practices in privacy policies and Google Play's data protection sections. Our findings reveal that 13 out of 15 applications (86.7%) leaked personal and device-related data—such as IP addresses, operating systems, and device identifiers—to third parties without obtaining parental consent, thereby breaching the General Data Protection Regulation (GDPR). Furthermore, several applications exhibited discrepancies between their stated privacy policies and their actual data collection behaviour. This research highlights the urgent need for stricter regulatory enforcement and more transparent software development practices to protect children's privacy in the digital ecosystem [5].



4.5.2 objective

The objective of this study is to investigate the privacy practices of popular children's mobile applications by analysing their network traffic, identifying the extent of personal data leakage to third parties, and evaluating whether these practices comply with GDPR requirements and align with their stated privacy policies and data protection disclosures. [5]

4.5.3 Methods

This study employed a three-phase methodological approach to evaluate the privacy practices of children's mobile applications. First, fifteen applications were selected from the Google Play Store's "Kids" section, representing three age categories (0–5, 6–8, and 9–12 years), using a saturation-based selection strategy to ensure a representative sample. Second, network traffic generated by these applications was recorded using an Android device connected through a Linux-based access point. Tools such as *mitmproxy* and *top dump* were utilized to intercept, log, and inspect data packets transmitted between the applications and external servers while actively testing their main functionalities. Third, the captured data was analyzed to identify instances of personal data transfer, including IP addresses, device identifiers, timestamps, and user information, and then cross-referenced with each application's privacy policy and Google Play's Data Protection section to assess transparency, disclosure accuracy, and compliance with GDPR requirements. [5]

4.5.4 Result

The analysis revealed a significant prevalence of undisclosed personal data sharing among the examined children's mobile applications. Out of the fifteen analyzed apps, thirteen (86.7%) transmitted personal or device-related information—such as IP addresses, phone models, operating system details, and unique identifiers—to third-party services without parental consent, in direct violation of GDPR requirements. The most frequently leaked data items were IP addresses and device characteristics, while more sensitive data, including a child's first name, email address, and even microphone or camera usage, were detected in isolated cases. Furthermore, a substantial discrepancy was found between the recorded network traffic and the information disclosed in both the privacy policies and the Google Play Data Protection sections of these applications; several apps falsely claimed not to collect or share personal information despite evidence to the contrary. Only two applications refrained from transmitting personal data entirely, indicating that privacy-preserving practices remain the exception rather than the norm in this category. [5]

4.6 “Child Tracking using IoT Device”

4.6.1 Abstract

IoT-based child tracking has drawn a lot of interest as a way to guarantee children's protection and safety in a variety of settings. A GPS module, a microprocessor, and wireless communication modules are included in the suggested system. The microcontroller manages and analyses the information after the GPS module has collected the child's location data. This platform, which is accessed through a smartphone application, enables parents or guardians to track the whereabouts of their children in real time. The GPS module is used by the child tracking system to continuously collect the child's position data. Real-time tracking is made possible by the platform, which receives and maintains the location data. Parents or guardians can use the smartphone application to track the child's location on a map. They are able to create predetermined safe zones and get alerts any time the child enters or exits them. The Internet of Things based child tracking system offers a complete solution that enhances child safety and gives parents and other carers peace of mind.[6]



4.6.2 Objective

The objective of this study is to design and implement an IoT-enabled child tracking system using a GPS module and NodeMCU microcontroller to provide real-time location monitoring, enable safe zone configurations, and deliver instant alerts to enhance children's safety and give peace of mind to parents and guardians [2].

4.6.3 Methods

The methodology of this research involved the development and testing of a GPS and IoT-based tracking system. A NodeMCU microcontroller was integrated with a NEO-6M GPS module to continuously capture the latitude and longitude of the device. Data was transmitted through a Wi-Fi network to a SQL Server database, from which an Android mobile application retrieved the information and displayed it on Google Maps. The system allowed parents to set radius-based geofencing alerts, which triggered notifications when the child crossed predefined boundaries. Hardware connections were established between the GPS module and the NodeMCU, and the software implementation was carried out using Arduino IDE and Blynk platform for real-time monitoring [6].

4.6.4 Result

The proposed system successfully demonstrated its ability to track a child's location in real time and transmit accurate latitude and longitude coordinates to a server. Parents were able to view the current position of their child on Google Maps, including direction and speed, and received alerts when the child moved beyond preset safety zones. The device operated efficiently with low power consumption (approximately 45mA during active tracking) and supported wide-area connectivity through Wi-Fi networks. Experimental testing confirmed that the system provided reliable and prompt notifications, offering an effective solution for child safety monitoring [6].

4.7 “Developing an early literacy mobile app: design process and testing the app with children with special needs”

4.7.1 Abstract

Early literacy skills are vital for later reading success, and technology-based interventions have shown potential in supporting these skills. This study developed a mobile application aimed at improving early literacy, particularly phonological awareness, for children with special needs. Using a design and development research model, the study included focus groups, a pilot trial, and an experimental intervention with 32 children (16 with special needs and 16 typically developing). Findings indicated that the app enhanced phonological awareness among children with special needs, though their scores remained lower than typically developing peers. Parents and teachers valued the app for its usability and content but also raised concerns about screen time. Overall, integrating educational technologies into literacy-focused lesson plans can enhance instructional quality and outcomes.[7].



4.7.2 Objective

The study aimed to design and evaluate a mobile application that supports the development of early literacy skills in children with special needs. Specifically, the objectives were: (1) to collect feedback from teachers and mothers regarding the app’s usability, features, and potential improvements, and (2) to test whether the app significantly improves phonological awareness performance in children with special needs when integrated into lesson plans.[7].

4.7.3 Methods

The research followed a design and development research model involving multiple phases. Focus group discussions with experts in early literacy and educational technology guided the app’s design. A pilot study with children, mothers, and teachers provided further feedback for revisions. Finally, an experimental study was conducted with 32 children (16 in the experimental group—children with autism, Down syndrome, or hearing impairment, and 16 in the control group—typically developing peers). Children in the experimental group participated in 10–14 sessions using the mobile app, while the control group continued standard preschool education. Pre- and post-test assessments were conducted using the phonological awareness subtest of the Test of Early Literacy (TEL). Thematic analysis was applied to interview data from teachers and mothers, while quantitative results were analyzed using non-parametric tests.[7].

4.7.4 Results

The qualitative findings revealed three main themes: Content & Usability, Visual Features, and Concerns & Suggestions. Teachers and mothers highlighted the benefits of age-appropriate, adaptive content and high-quality visuals, while expressing concerns about screen time and skill generalization. The experimental results showed that children with special needs in the experimental group significantly improved their phonological awareness scores after using the app (mean pre-test = 2.87; mean post-test = 8.43). In contrast, the control group showed no significant difference between pre-test and post-test scores. Despite improvements, the post-test scores of children with special needs remained lower than those of typically developing peers. Teachers also reported that the app could be effectively integrated into lesson plans, saving preparation time and motivating students.[7].

4.8 "Harnessing Mobile Health Technologies to Improve Child Health Outcomes: A Systematic Review of Interventions for Under-5 Year Children in Low- and Middle-Income Countries"

4.8.1 Abstract:

This systematic review assesses the efficacy of mobile health (mHealth) technologies in enhancing the health outcomes of children under five in low- and middle-income countries (LMICs). From an initial pool of 518 studies, 22 were selected, highlighting mHealth's role in boosting immunization rates, supporting exclusive breastfeeding, monitoring child growth, and improving maternal health-seeking behaviors. Text messaging emerged as a cost-effective tool for behavior change and healthcare utilization, offering a promising solution to address healthcare gaps in resource-limited settings [8].



4.8.2 Objectives:

The study aims to evaluate the effectiveness of mHealth interventions in improving early detection, prevention, and management of common childhood illnesses among children under five in LMICs. It focuses on increasing immunization coverage, promoting exclusive breastfeeding, enhancing growth monitoring, and fostering better health-seeking behaviors among mothers, with the goal of informing policy and resource allocation in these regions [8].

4.8.3 Methods :

The review employed a comprehensive literature search across databases like PubMed, CINAHL, SCOPUS, Science Direct, and PsycINFO, covering English-language studies from 2013 to 2024, supplemented by grey literature and snowballing techniques. Of 518 identified studies, 440 were excluded based on abstracts and titles, 29 after full-text review, leaving 22 studies—17 randomized controlled trials, three quasi-experimental, one descriptive, and one methodological—for analysis following PRISMA guidelines [8].

4.8.4 Results:

Findings from 22 studies, predominantly from Sub-Saharan Africa and South Asia, showed mHealth significantly improved child health outcomes. Eleven studies reported higher immunization rates (e.g., 97% vs. 82% at six weeks in Zimbabwe, $p < 0.001$), six enhanced exclusive breastfeeding (e.g., 16% increase in Bangladesh, RR:1.16), two improved growth monitoring, and three boosted maternal health-seeking behaviors. Text messaging and voice reminders were key, with methodological quality rated as good (6-8 on PEDro scale) across most studies [8]

4.9 Parenting Apps for Children and Service Review "

4.9.1 Abstract:

This systematic review examines parenting apps available on Google Play and Apple App stores, targeting parents of children aged 2-12, to summarize their characteristics and identify improvement needs. From 522 apps, 53 met criteria for providing active interventions, mostly free (86.7%) but often requiring subscriptions (32%). Apps focused on child development assessment, rewards, and behavior tracking, drawing from behavioral therapy or positive psychology, but lacked personalization, mood tracking, and evidence from studies on their effects [9].



4.9.2 Objectives:

The review aims to explore the range of parenting apps accessible via major platforms, assess their evidence-based status, and describe key features and tools. It seeks to highlight how apps can overcome barriers to traditional parenting programs (e.g., time, stigma) while identifying gaps like lack of personalization and psychoeducation to guide better app development for parents needing support in child-rearing practices[9].

4.9.3 Methods:

A systematic search was conducted in June 2021 on Google Play and Apple App stores using terms like "parent," "parenting," and "program." After removing duplicates (from 522 to 355 apps), eligibility was assessed in two steps: platform descriptions/screenshots, then in-app testing. Inclusion required English-language apps targeting parents of 2-12-year-olds with active interventions (e.g., child-focused activities). Two reviewers independently coded features; disagreements were resolved by a third. Academic databases were searched for app efficacy studies [9].

4.9.4 Results:

Of 53 included apps, most fell under "parents" (37.7%) or "lifestyle/education" categories (22.6% each), with ratings averaging 4 stars but low review rates (0.31-13%). Features included text (100%), images/animations (54.7-45.3%), child profiles (62.3%), rewards (47.2%), and goal trackers (37.7%), but rarely personalization (1.9%) or mood tracking (1.9%). Techniques drew from behavioral therapy (39.6%) and positive psychology (18.9%), with no apps having efficacy studies [9].

4.9.5 Problem & Solution:

Problems: Many apps lack evidence-based validation, personalization, psychoeducation, and online communities, leading to potential inefficacy; high heterogeneity in quality and low review engagement make selection challenging for parents.

Solutions: Anchor apps in proven frameworks like CBT, add transdiagnostic techniques and personalization via user data, include psychoeducation and communities for engagement, and conduct clinical trials to ensure effectiveness and cultural relevance [9].

4.10 Enhancing parent-child interaction through the gamification of parenting: An exploration of digital interventions"

4.10.1 Abstract:

This systematic review of studies from 2010 to 2023 on the impact of digital devices on children's behavior and academic achievement, focusing on the role of parents in gamification to promote positive interaction. It highlights positive aspects, such as improved cognitive skills through educational games, and negative aspects, such as aggression and emotional problems, from excessive use, proposing strategies for balancing use [10].



4.10.2 Objectives:

The review aims to explore the range of parenting apps accessible via major platforms, assess their evidence-based status, and describe key features and tools. It seeks to highlight how apps can overcome barriers to traditional parenting programs (e.g., time, stigma) while identifying gaps like lack of personalization and psychoeducation to guide better app development for parents needing support in child-rearing practices [10].

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Problems: Many apps lack evidence-based validation, personalization, psychoeducation, and online communities, leading to potential inefficacy; high heterogeneity in quality and low review engagement make selection challenging for parents.

Solutions: Anchor apps in proven frameworks like CBT, add transdiagnostic techniques and personalization via user data, include psychoeducation and communities for engagement, and conduct clinical trials to ensure effectiveness and cultural relevance [10].

4.11 “Arabic apps for Baby care paper”

4.11.1 Abstract

This study aims to assess the needs and requirements for a new, comprehensive well-baby clinic mobile application and to develop a prototype that covers all service components: physical growth, developmental milestones, immunizations, nutrition, and safety & emergency measures. [11]



4.11.2 Problem

The study indicates that the use of healthcare services for children under five in Egypt is considered low. Well-baby visits are mainly for immunization in the first two years of age. Additionally, parents may rely on the internet and mobile applications for support in day-to-day parenting situations. The researchers found that the available Arabic applications on the market do not provide a comprehensive approach to well-baby clinic services on a scientific basis. [11]

4.11.3 Solution

The researchers developed a novel well-baby clinic app called

"Sehhat Tefly". This app aims to provide solutions to the identified problems by:

- **Providing Comprehensive Content:** The app includes services for physical growth, developmental milestones, immunizations, nutrition, and safety & emergency measures, relying on evidence-based information.
- **Offering Interactive Features:** Unlike the available apps, which were in the form of written articles, this app includes interactive features to help parents monitor their children's health.
- **Adhering to Scientific Standards:** The app was developed by researchers from a university, ensuring the information is reliable and scientific, unlike commercial apps that include ads.
- **Meeting User Needs:** An online survey of 374 mothers showed that almost all (97.5%) reported the need for a new app to regularly follow their children's health. [11]

4.11.4 Result

Thirteen Arabic apps were downloaded and reviewed. All of these apps (100% or 13/13) were in the form of written articles that offered no interactive features for parents. They all offered general health education not tailored to the status of the child. None of them were developed by a university or a certified health organization. Almost all (92.3% or 12/13) were commercial apps as they included ads. Our online survey of 374 mothers showed that almost all of them (97.5%) reported the need for a new app to regularly follow their children's health. They also reported the need for nutrition, developmental milestones, and physical growth services in the app. We developed the "Sehhat Tefly" app, which includes caregiver information, child sociodemographic characteristics, and the six previously mentioned services. It also included health education videos and a conclusive report section. [11]

4.12 "App-based learning for kindergarten children at home"

4.12.1 Abstract

The study's primary objective is to develop and test computer tablet applications ("apps") to determine if they can help parents create high-quality home learning environments (HLEs) and positively influence children's short- and long-term development of literacy and mathematical skills. The "App-based learning for kindergarten children at home" (Learning4Kids) project is a five-year, longitudinal study funded by the EU, designed to evaluate the impact of a tablet-based family intervention on children's learning, development, social inclusion, and well-being. [12]



4.12.2 Problem

Many children, especially those from low socioeconomic status (SES) backgrounds, are at risk of not reaching sufficient competence levels in literacy and mathematics because they may lack the necessary home support. While formal teaching begins in primary school, children acquire important skills for later learning long before they start school. Research indicates that a child's family background, specifically their SES and the quality of their home learning environment (HLE), significantly impacts their development of these competencies. However, many existing educational apps do not use developmentally appropriate learning practices and have not been evaluated for their impact on learning outcomes, making it difficult for parents to find effective apps. [12]

4.12.3 Solution

The study aims to use new technological possibilities offered by tablet computers to improve the HLE and support children's competencies.

The Learning4Kids project plans to address this problem by:

- **Developing and using high-quality apps:** The project will identify and develop innovative apps that provide information on the HLE, offer effective math games and children's books, and encourage family participation in the intervention. The apps will be designed based on the "four pillars" of active, engaged, meaningful, and socially interactive learning, which are key characteristics of efficient educational apps.
- **Implementing a family intervention approach:** The study will use a non-intensive, easy-to-apply intervention to improve the HLE and influence children's long-term learning trajectories. This approach will focus on directly stimulating children, educating parents, and enriching parent-child interactions and the caregiving environment.
- **Targeting at-risk families:** The project will specifically aim to recruit and retain families with low SES and/or a migration background, as these children are at a greater risk of academic failure. The study will use an inclusive recruitment plan, including providing information in multiple languages to reach diverse families. [12]

4.12.4 Result

This article is a study protocol and does not present the final results of the intervention. It outlines the methods and design for the first cohort of the study.

- The study uses a multi-method intervention approach, drawing on expertise from psychology, education, informatics, and didactics.

- Data will be collected via standardized tests, observations, and parental and educator surveys and checklists. App usage times will be assessed through mobile sensing.
- The first cohort of the sample consists of 190 families assigned to one of four groups: a business-as-usual group, two intervention groups (one for mathematical apps and one for literacy apps), and a tablet-control group.
- The study protocol also details the specific tests and surveys used to assess children's mathematical, linguistic, and cognitive competencies, as well as the home learning environment and family background. [12]

4.13 "Parenting apps review: in search of good quality apps"



4.13.1 Abstract

This paper presents a review of high-quality parenting apps available to parents and healthcare professionals. First-time parents often use mobile applications to help them adjust to their new roles. However, the large number of low-quality apps makes it difficult to find reliable ones. The study evaluated 16 apps for their quality using the Mobile App Rating Scale (MARS). [13]

4.13.2 Problem

New parents, especially first-time parents, often feel unprepared for their new roles and rely on the internet and mobile applications for support. While there are many apps available to help with parenting, a significant number of these apps are of low quality and can contain inaccuracies, irrelevant information, and frustrating advertisements. This makes it difficult for parents and healthcare professionals to find reliable and useful apps. [13]

4.13.3 Solutions

The researchers searched the Google Play Store on June 1, 2018, using 18 different search terms related to parenting. Of the 4,300 free apps identified, 16 met the eligibility criteria. These 16 apps were then evaluated using the Mobile App Rating Scale (MARS), a tool designed by a team of healthcare professionals and app developers. The MARS tool evaluates apps on four subscales: engagement, functionality, aesthetics, and information. [13]

4.13.4 Results

Of the 4,300 apps initially found, only 16 met the quality criteria for review, representing just 0.4%. All 16 apps were free, publicly available, and had a privacy policy. Twelve of the apps were updated within the last year, and eight were free of advertisements. The apps were divided into categories: tracking apps, information apps, sleeping-aid apps, photo-sharing apps, and miscellaneous apps. [13]

- **Overall MARS Scores:** Four apps received the highest MARS score of 4.5/5, and one received the lowest score of 4/5. The remaining 11 apps scored between 4.2 and 4.4 out of 5.
- **Top-Rated Apps by Category:**
 - Tracking: Child Growth Tracker (4.5/5).
 - Information: Baby and Child First Aid (4.5/5).
 - Sleeping-aid: Don't Cry My Baby (Lullaby) (4.5/5).
 - Photo sharing: FamilyAlbum - Easy Photo & Video Sharing (4.5/5).

4.14 "Smart Sleep Monitoring: An Integrated Application for Tracking and Analyzing Babies' Sleep"

4.14.1 Abstract and Introduction



This paper introduces an innovative application named: Baby Care, designed to help parents monitor and analyze their children's sleep patterns. The application integrates hardware sensors with a software solution to capture, process, and securely store sleep data in a cloud database. The key features include real-time monitoring, historical data visualization, and alert notifications for abnormalities. The goal is to provide a comprehensive and user-friendly tool that gives parents peace of mind about their child's health and development. The application focuses on monitoring vital parameters like : oxygen levels (SpO2) and heart rate, which are crucial for detecting conditions such as sleep apnea and respiratory distress. [14]

4.14.2 Comparison with Existing Solutions

Before developing BayCare, the researchers analyzed existing products on the market, noting their strengths and weaknesses.

- Angelcare AC127: Monitors breathing and movement using sensors under the mattress and includes a camera for remote observation, but it lacks a mobile application and the ability to track historical data.
- Nanit Complete: Offers high-quality camera resolution and tracks breathing and growth, but requires a subscription, has a complex installation, and its fixed stand makes it impractical for use outside the home.
- Owlet Smart Sock: Uses a sock with sensors to track vital signs, with data displayed in a mobile application. However, it is expensive and has limited functionality. Some users reported the sock can be uncomfortable for the baby and the system doesn't provide an option to contact a doctor in case of issues. [14]

4.14.3 Baby Care Application Features

The Baby Care application was designed to address the shortcomings of existing solutions. Its key objectives and features include:

- User-Friendly Interface: An intuitive interface was designed for both Android and iOS platforms to display real-time and historical data on sleep and vital signs.
- Monitoring Hardware: The system integrates a MAX30100 sensor to measure pulse and oxygen levels, communicating with an Arduino microcontroller and a Raspberry Pi.
- Cloud Integration: Data is processed and stored securely in a Firebase cloud database.
- Alerts and Notifications: The application alerts parents to unusual or alarming values and provides recommendations for next steps. When location services are enabled, it can even find the nearest medical help.
- User Accounts: Users can create personal accounts, add multiple children, and customize profiles.
- Weight Tracking: The app allows parents to track and visualize their child's weight development over time.

The BabyCare system is designed with an open-source architecture that is easily expandable and supports edge computing capabilities. The project is currently a prototype with a focus on functionality rather than final aesthetics. [14]

4.15 "Lifestyle Spectra for Children with Autism: Using Device Applications to Develop Some Skills"

4.15.1 Abstract

The research aimed to study the effectiveness of using smart device applications in developing some life skills for children with autism spectrum disorder. The study used an experimental approach with a single-group pre- and post-measurement design. The sample consisted of 10 children with autism spectrum disorder, aged 5-7 years [15].



The research tools included the Gilliam Autism Rating Scale, the Stanford-Binet Intelligence Scale (Fifth Edition), a Life Skills Scale (prepared by the researcher), and a smart device application program for developing some life skills for children with autism spectrum disorder (prepared by the researcher).

The results showed statistically significant differences at a significance level of 0.01 between the children's scores in the experimental group before and after the program application on the life skills scale in the direction of post-measurement. This indicates a decrease in the children's scores and thus an improvement in their skills after exposure to the program's content. The results also showed no statistically significant differences between the post- and follow-up measurements, which means the improvement was sustained over the follow-up period [16].

4.15.2 Problems

The research problem was crystallized from the researcher's experience as a special education specialist. The researcher observed that most children with autism spectrum disorder suffer from a deficit in multisensory information processing, participation, and life skills [15]. This leads to an inability to independently perform daily tasks and function successfully [15].

This deficit affects the children's performance in daily life skills when compared to their non-autistic peers who match their intelligence levels [18]. Children with autism also suffer from qualitative deficits in life skills, which is clearly seen in their lack of awareness of others and failure to respond to external stimuli [16]. The study aims to fill a gap in this field by focusing on how to develop some life skills through a program based on smart device applications. The problem is summarized in the main question: "What is the effectiveness of using smart device applications in developing life skills for children with autism spectrum disorder?"

4.15.3 Solutions

The research proposed a program based on smart device applications to develop life skills for children with autism spectrum disorder. The use of smart device applications serves as an effective and precise tool for developing life skills support programs. The technological advancements in computer applications and smart devices have provided educational tools for children with disabilities. The research literature identified several justifications for using these applications with children with autism spectrum disorder, including: [15]

- **Attracting Attention:** They attract the attention of children with autism and reduce their chances of being distracted [20, 21].
- **Supporting Daily Skills:** They support daily life skills and other aspects of development [15].
- **Providing a Clear Structure:** They provide a clear structure and expectations for skill development [15].

The proposed solution includes a set of activities, skills, games, and plans organized through educational videos and games designed to run on smartphones and tablets. These applications assist the child in successfully processing and developing the life skills they use in their daily lives. These skills include basic tasks such as:

- Washing hands before and after eating [5].
- Making a sandwich
- Using a towel.
- Using the bathroom
- Getting dressed.

The program also aims to assist parents in training their children using smart devices to develop attention and life skills at home. This approach helps the child acquire more interactive skills (verbal, non-verbal, and social) to modify their behavior.

4.15.4 Results

The research results revealed statistically significant differences at a significance level of (0.01) between the scores of children in the experimental group before and after the program implementation on the life skills scale, in the direction of the posttest. This indicates a decrease in the scores of children in the experimental group and, consequently, an improvement after exposure to the program content. The results also showed no statistically significant differences between the posttest and follow-up test scores, indicating that the improvement continued throughout the follow-up period.

4.16 *"Requirements for using some Artificial Intelligence (AI) Applications to develop creative thinking skills among Kindergarten children as perceived by kindergarten teachers"*

4.16.1 Abstract

This study aimed to identify the effectiveness of using artificial intelligence (AI) applications and the development of creative thinking skills in kindergarten children from the perspective of kindergarten teachers [3, 4]. It also aimed to explore the relationship between the dimensions of AI applications—after training and development, after suitability, and after effectiveness—and the development of creative thinking skills [16].



The study used a descriptive-analytical approach. The study sample consisted of (10) kindergarten teachers in the city of Jeddah, Saudi Arabia [16].

4.16.2 Problems

The problems addressed by the research are as follows:

- **Lack of Awareness of AI Benefits:** Kindergarten teachers face challenges in understanding how to effectively use AI applications in the educational process and how to employ them to develop creative thinking skills in children
- **Lack of Suitability of Some Applications:** Some currently available applications may not be specifically designed to meet the needs of preschool children, which limits their effectiveness in achieving desired educational goals .
- **Need for Training:** Teachers require specialized training to enhance their skills in using these applications correctly and innovatively, in a way that serves the development of children's creative thinking
- **Lack of Effectiveness:** Some applications face problems in achieving the required effectiveness in skill development due to weak interaction or content that is incompatible with the characteristics of children's growth [16].

4.16.3 Solutions

The research provides a set of solutions to address the mentioned problems, which are:

- **Professional Training and Development:** Training programs must be provided for teachers to introduce them to the latest AI applications and how to use them in a systematic and innovative way
- **Developing Suitable Applications:** AI applications should be developed specifically to suit the age group of kindergarten children, focusing on interactive interfaces, attractive colors, and enjoyable content
- **Focus on Effectiveness:** Applications must focus on achieving clear and specific educational goals, and their effectiveness should be evaluated periodically to ensure the desired results are achieved
- **Integration with the Curriculum:** These applications can be integrated as an essential part of the curriculum, allowing teachers to use them regularly to support the learning process and develop creative thinking

4.16.4 Results

The study's results revealed the following:

- A strong relationship exists between the use of AI applications and the development of creative thinking skills in kindergarten children
- A statistically significant relationship exists between the dimensions of AI applications (training and development, suitability, effectiveness) and the development of creative thinking skills in kindergarten children
- No statistically significant differences were found between the teachers' responses regarding the requirements for using AI applications to develop creative thinking skills in kindergarten children, which are attributable to the variables of years of experience or academic qualification
- The results confirmed that AI applications can be an effective tool in supporting the educational process, especially in developing higher-order thinking skills in children

4.17 Research Papers on Children Applications and Technologies

<i>Paper Title</i>	<i>Year</i>	<i>Objective</i>	<i>Technology Used</i>	<i>Advantages</i>	<i>Disadvantages</i>	<i>Metrics</i>	<i>Dataset Used</i>
Arabic Children's Applications on (Android): An Analytical Study of Usage Trends	2022	Analyze Arabic apps for children on Android to study their impact on cultural identity and knowledge.	Android mobile applications (Google Play).	• Provides solutions for improving app quality and enhancing cultural content. • Highlights need for cooperation between institutions and parents.	• Weak/scarcely Arabic content. • Dominant foreign cultural influence. • Lack of interaction in most Arabic apps. • No institutional support.	• Download counts (e.g., 1.2M for educational apps). • Content distribution % (e.g., 28.5% Games). • 40% improvement in Arabic language for users.	Arabic applications on the Google Play store.
Quality of Mobile Apps for Child Development Support: Search in App Stores and Content Analysis	2022	Examine the quality of mobile apps designed to support parents in promoting child health and development.	Mobile applications (Google Play and Apple App Store).	• Generates insights to guide future developers in creating evidence-based, culturally responsive, and user-friendly apps.	• Most apps offer generic advice. • Lack personalization, cultural sensitivity, and evidence-based grounding. • High user burden (costs, privacy issues).	• 8.7% of apps referenced evidence-based guidelines. • User Burden Scale. • Bright Futures Guidelines.	149 mobile apps from Google Play & Apple App Store.
The effectiveness of early childhood development programs	2023	Systematically review the effectiveness of early childhood development (ECD) programs on outcomes for disadvantaged children.	Not technology-focused; reviews ECD programs like Head Start.	• ECD programs significantly improve school readiness, academic achievement, and IQ. • Reduce grade retention and special education placement.	• Insufficient evidence for long-term social, health, and family outcomes. • Many children start school without adequate readiness skills.	• Cognitive outcomes (IQ). • Educational outcomes (grade repetition). • Social outcomes (graduation rates).	Systematic review of studies on ECD programs for children aged 3–5.
wide scale well-baby clinic mobile application	2020	Introduce "Sehhat Tefly," an mHealth app in Egypt to help caregivers monitor the health and wellbeing of children under five.	Mobile health (mHealth) application ("Sehhat Tefly") on Android.	• App is feasible, accessible, free, interactive, and evidence-based. • Complements traditional clinic services effectively.	• Traditional clinics are underutilized. • Existing Arabic apps are limited, ad-driven, and lack scientific backing from health organizations.	• Mobile App Rating Scale (MARS) score of 3.7/5. • 1,445 downloads and 943 user registrations in pilot.	Pilot study data from the "Sehhat Tefly" app.
Privacy in Popular Children's Mobile Application A Network	2020	Investigate privacy practices of children's apps by analyzing network traffic	Network traffic analysis tools (mitmproxy, top dump)	• Highlights the urgent need for stricter regulatory enforcement	• 86.7% of apps leaked personal data without consent, breaching	• 13 out of 15 apps (86.7%) leaked personal data.	15 popular children's apps from the Google Play Store.

Traffic Analysis		for data leakage and GDPR compliance.	on Android mobile applications.	and developer transparency to protect children's privacy.	GDPR. • Major discrepancies found between privacy policies and actual data collection.		
Child Tracking using IoT Device	2023	Design and implement an IoT-enabled child tracking system to provide real-time location monitoring and safety alerts.	IoT device with GPS module (NEO-6M), NodeMCU microcontroller, Wi-Fi, SQL Server DB, and an Android app.	• Provides reliable, real-time tracking and safe zone alerts. • Offers peace of mind to parents. • Low power consumption.	• Implied problem is the lack of effective, real-time child safety monitoring tools for parents.	• Power consumption of approx. 45mA during active tracking.	Experimental testing of the developed prototype system.
Developing an early literacy mobile app: design process and testing the app with children with special needs	2017	Design and evaluate a mobile app to support the development of early literacy skills (phonological awareness) in children with special needs.	Custom-developed mobile application for literacy.	• Significantly enhanced phonological awareness in children with special needs. • App was valued by parents/teachers for its content and usability.	• Users with special needs still scored lower than typically developing peers. • Parents raised concerns about screen time.	• Test of Early Literacy (TEL) pre- and post-test scores (mean improved from 2.87 to 8.43).	32 children (16 with special needs, 16 typically developing)
Harnessing Mobile Health Technologies to Improve Child Health Outcomes...	2013–2024	Assess the efficacy of mHealth technologies in enhancing health outcomes for children under five in low- and middle-income countries (LMICs).	Mobile health (mHealth) tech, mainly text messaging (SMS) and voice reminders.	• mHealth is a cost-effective tool that significantly boosts immunization rates, breastfeeding, and health-seeking behaviors.	• Addresses the problem of significant healthcare gaps in LMICs for children under five.	• Immunization rates (e.g., 97% vs. 82%). • 16% increase in breastfeeding. • PEDro scale for quality (6-8).	Systematic review of 22 studies from health databases.
Parenting Apps for Children and Service Review	2021	Summarize the characteristics of available parenting apps (for ages 2-12) and identify key areas for improvement.	Parenting mobile applications (Google Play and Apple App stores).	• Apps can help overcome barriers to traditional parenting programs (e.g., time, stigma). • The review guides better app development.	• Apps lack personalization, mood tracking, and evidence-based validation. • High heterogeneity in app quality.	• App store ratings. • Feature percentages (e.g., only 1.9% offered personalization).	53 parenting apps selected from 522 apps on app stores.
Enhancing parent-child interaction through the gamification of parenting...	2010–2023	Explore the impact of digital devices on children's behavior and academic achievement, focusing on gamification.	Digital devices and educational games.	• Educational games can improve children's cognitive skills.	• Excessive use can lead to aggression and emotional problems. • Lacks personalization and evidence-based validation.	(Results in source text are duplicated from paper 4.9)	Systematic review of studies from 2010 to 2023.

Arabic apps for Baby care paper	2020	Assess needs for and develop a prototype of a comprehensive well-baby clinic mobile application ("Sehhat Tefly").	Mobile application ("Sehhat Tefly").	• The proposed app is comprehensive, interactive, and based on scientific, evidence-based information. • Meets user-reported needs.	• Existing Arabic apps are non-interactive, commercial (with ads), and not from certified health organizations.	• 100% of reviewed apps were non-interactive. • 97.5% of surveyed mothers reported needing a new, better app.	Review of 13 existing Arabic apps and a survey of 374 mothers.
App-based learning for kindergarten children at home	2018	Develop and test if tablet apps can help parents create high-quality home learning environments and improve children's skills.	Tablet-based educational applications.	• The proposed approach can improve the home learning environment and support children's competencies, especially for at-risk families.	• Many children, especially from low-SES backgrounds, lack home support. • Existing apps are often not developmentally appropriate.	(Study protocol, no final results). Planned metrics include standardized tests, surveys, and mobile sensing.	(Study protocol). Proposed sample is 190 families for the first cohort.
Parenting apps review: in search of good quality apps	2018	Review high-quality parenting apps to help parents and healthcare professionals find reliable ones from a crowded market.	Parenting mobile applications (Google Play Store).	• Identified and categorized 16 high-quality apps using a structured quality scale, providing a useful guide for parents.	• The vast majority of parenting apps are low-quality, making it hard to find reliable ones (only 0.4% met criteria).	• Mobile App Rating Scale (MARS) scores (top apps scored 4.5/5).	16 apps selected from an initial pool of 4,300 on the Google Play Store.
Smart Sleep Monitoring Integrated Application for Tracking and Analyzing Babies' Sleep	2019	Introduce the "Baby Care" app, which uses integrated hardware and software to help parents monitor and analyze their children's sleep patterns.	Mobile app ("Baby Care"), hardware sensors (MAX30100), Arduino, Raspberry Pi, Firebase DB.	• Addresses shortcomings of existing products by offering historical data, alerts, and an open-source architecture.	• Existing commercial solutions are expensive, complex, lack key features, or can be uncomfortable for the baby.	• Monitors vital parameters like oxygen levels (SpO2) and heart rate.	Prototype development and comparison against 3 existing market products.
Lifestyle Spectra for Children with Autism: Using Device Applications to Develop Some Skills	2021	Study the effectiveness of using smart device applications to develop essential life skills for children with autism.	Custom-developed smart device application with educational videos and games.	• App-based program led to statistically significant and sustained improvement in the life skills of children with autism.	• Children with autism often have deficits in life skills, preventing them from performing daily tasks independently.	• Statistically significant differences ($p < 0.01$) between pre- and post-test scores on a Life Skills Scale.	10 children with autism spectrum disorder, aged 5-7 years.

Requirements for using some Artificial Intelligence (AI) Applications to develop creative thinking skills among Kindergarten children	Not Specified	Identify the effectiveness of AI applications in developing creative thinking skills in kindergarteners, from a teacher's perspective.	Artificial Intelligence (AI) applications.	• A strong, statistically significant relationship exists between using AI apps and developing creative thinking in children.	• Teachers lack awareness of AI benefits. • Available apps are often not suitable for kindergarteners. • Teachers require specialized training.	• Statistically significant relationship between AI app dimensions (training, suitability, effectiveness) and creative thinking.	Survey of 10 kindergarten teachers in Jeddah, Saudi Arabia.
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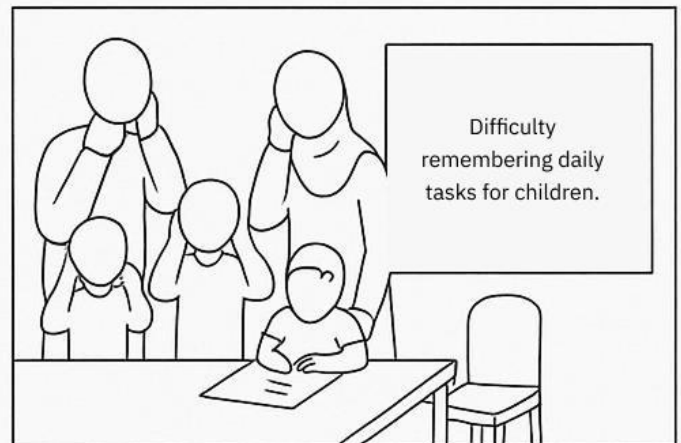
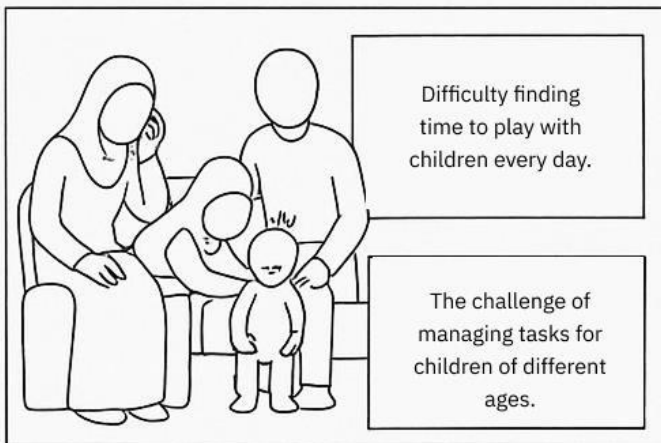
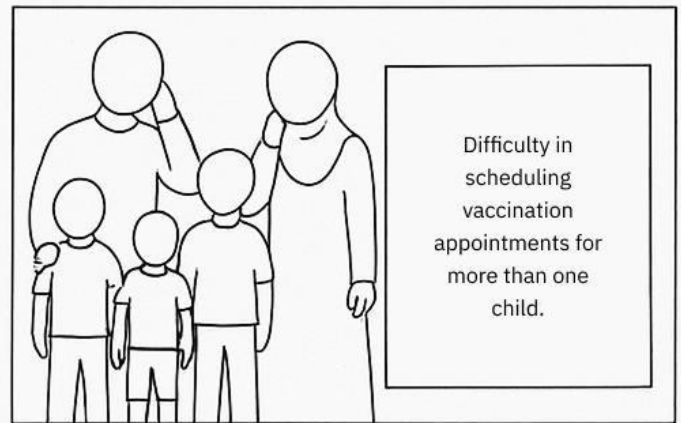
4.2 Analysis of Existing Solutions and Platforms (Competitive Analysis)

To determine our project's position in the market and ensure the provision of a unique value, by the grace of God, a comprehensive analysis was conducted of the existing solutions and platforms in the field of child care. This analysis aims to understand competitors' strategies, evaluate their strengths and weaknesses, and discover the gaps in the market that our project can fill. By studying the features and services offered by other applications, we can build a program that does not repeat the available solutions, but rather develops them and provides a more integrated and effective experience that meets the needs revealed by user research.

4.2.1 Analysis table: (Table No. 1)

COMPARISON ELEMENT X ✓	BABY : YOUR BABY TRACKER	BRIGHTWHEEL: CHILDCARE APP	BABY TRACKER
Vaccination Reminders	✓	X	✓
Vaccination Guidelines	✓	X	X
Vaccination Locations	X		X
Feeding Schedule	✓	X	✓
Diet	✓	-	✓
Growth Index	✓	✓	X
Nutrition Statistics	✓	X	X
Educational Content	✓	✓	X
Educational Stories	X	X	X
Mind Games	X	✓	X
Learning Rewards	X	✓	X
Play Levels	X	X	X
Study Materials	X	X	X
Educational Challenges	X	✓	X
Learning Alerts	X	✓	X
First Aid	✓	X	X
Healthy Community	✓	-	X
Medical Consultations	✓	X	X
Medical Guide	X	✓	X
Sleep Content	X	X	X
Sleep Organization Solutions	X	X	X
Sleep Log	X	X	✓
Child Sleep Quality Analysis	-	✓	X
Sleep Routines	X	X	X
Psychological Centers	X	X	X
Educational Specialist	X	✓	X
Motivation Support	X	✓	X
Educational Training Scenarios	X	X	X
Behavioral Games	✓	X	X
Task Schedule	✓	✓	X
To-Do List	-	X	X

5.0 Storyboard



6.0 Proposed Solutions

Based on the results reached through user research and competitive analysis, this section aims to present the proposed solutions that have been designed specifically to meet the needs of parents and fill the gaps in the market. These solutions represent our vision for a comprehensive project that provides real value to the family and evolves with the child's growth. This part will present these solutions in an organized manner, clarifying how they address the challenges that parents face in each age stage.

Age Group (from one day - 1 year)

- **Problem:** Parents' anxiety about vaccinations and sometimes forgetting their appointments.
- **Solution:** Integrated vaccination services that include reminders, guidelines on side effects, and a list of nearby vaccination locations.
- **Problem:** The baby's sleep disorder and the inability to establish a fixed routine.
- **Solution:** Sleep routine tools that provide a log of sleep times, an analysis of patterns, and tips to create a comfortable daily routine.
- **Problem:** Difficulty in determining the child's nutritional needs and following their growth.
- **Solution:** Nutrition content and support that provides a nutrition schedule that changes with age, a growth indicator for the child, and health information about food.
- **Problem:** Fear of developing allergies when trying new foods.
- **Solution:** Health care services that provide information about allergy-causing foods.

Age Group (from 1 - 3 years)

- **Problem:** Behavioral challenges like stubbornness and sibling conflict.
- **Solution:** Educational and behavioral support that includes behavioral games and points for teaching how to handle emotions.
- **Problem:** Difficulty in dealing with the child's picky eating and refusal of certain types of food.
- **Solution:** Nutrition content and support that provides a varied dietary system and tips for dealing with the picky child.
- **Problem:** The desire to occupy the child's time with useful and age-appropriate content.
- **Solution:** Gamified Learning through educational stories, and simple mental games to develop skills.

Age Group (from 4 - 7 years)

- **Problem:** Parents' need for tools to follow the child's educational and behavioral progress.
- **Solution:** Gamified Learning with advanced levels, actual school materials, and challenges, with alerts for parents to follow performance.
- **Problem:** Difficulty in dealing with complex behaviors and special needs, such as severe tantrums or supporting children with special needs.
- **Solution:** Educational and behavioral support that provides training scenarios, an educational specialist, and dedicated sections for children with special needs.
- **Problem:** Difficulty in organizing daily tasks with multiple children.
- **Solution:** Daily task tools that allow setting a schedule to distribute tasks among the children.

• Business Model Canvas

Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
- Pediatricians, nutritionists, psychologists - Clinics & vaccination centers - Nurseries & early education centers - NGOs (child health/mental health) - Baby product brands	- App development & maintenance - Content creation (medical, nutrition, parenting scenarios) - AI/analytics for growth & sleep tracking - Community management - Partner onboarding	- One-stop newborn management app - Vaccination reminders + clinic locator - Personalized growth, nutrition, and sleep analytics - Parenting scenarios (interactive guidance) - Expert consultations (medical & educational) - Special needs & behavioral support - Parent community	- Personalized services (trackers & reminders) - Community-driven support (forums) - Expert-driven services (consultations, scenarios)	- New parents (esp. first-time, urban, middle-class) - Parents with toddlers (0–5 years) - Pediatric clinics & vaccination centers - Nurseries/early education centers
Key Resources	Channels	Cost Structure	Revenue Streams	
- Development team (mobile & backend) - Medical & educational content creators - AI/analytics models - Expert partnerships	- Mobile app (main platform) - Partnerships with hospitals, clinics, nurseries - Social media campaigns targeting new parents - In-app community & notifications	- App development & hosting - Content production (medical, nutrition, parenting) - Expert partnerships (consultants/trainers) - Marketing & user acquisition - Customer support & community moderation	- Freemium model (basic free, premium features paid) - Subscriptions (monthly/annual) - In-app consultations (commission) - B2B partnerships (clinics, nurseries) - Ads/sponsorships (baby brands)	

Step 1: Features Analysis

Core Business Model Features

- Vaccination reminders
- Vaccination guidelines
- Vaccination centers – possible partnership with clinics
- Nutrition planning & analytics – premium upgrade
- Growth tracker – freemium with advanced analytics
- Medical consultations – paid telemedicine
- Medical guide – free/basic, premium unlocks detailed
- Sleep management – freemium with advanced AI insights
- Data logging + routines – premium
- Educational specialist consultations – paid
- Psychological centers – partnerships/referrals
- Special needs support – niche value, could be premium
- Parenting scenarios – premium interactive content
- Parent community – engagement, but can host ads/sponsorships

Secondary / Value-Add Features

- Educational content
- Educational stories for kids
- Cognitive games
- Gamification, keeps kids engaged
- Early learning materials
- Learning notifications
- First aid info
- Behavioral tracking
- (Parent task management – retention)

Low Business Impact Features

- basic tips – easily available online
- basic preschool content – too much competition
- simple notifications – can be integrated with others

Step 2: Business Model Canvas

Customer Segments

- New parents (first-time especially, urban & middle-class)
- Parents with toddlers (0–5 years old)
- Pediatric clinics & vaccination centers (B2B partnerships)
- Nurseries/early education centers (potential expansion)

Value Propositions

- One-stop app for managing a newborn's health, nutrition, sleep, and development.
- Reliable vaccination reminders + location of clinics.
- Personalized growth, nutrition, and sleep analytics.
- Access to expert consultations (medical + educational).
- Parenting scenarios for real-life guidance.
- Special needs & behavioral support.
- Community for parent-to-parent connection.

Channels

- Mobile app (main platform).
- Partnerships with hospitals, clinics, nurseries.
- Social media campaigns targeting new parents.
- In-app community & notifications.

Customer Relationships

- Personalized (growth/sleep/nutrition trackers).
- Community-driven (peer support forum).
- Expert-driven (paid consultations, parenting scenarios)

Revenue Streams

- Freemium model → basic trackers & reminders free, premium for advanced analytics, parenting scenarios, sleep solutions.
- Subscription plans → monthly/annual for premium content.
- In-app consultations → commission per medical/educational session.
- B2B partnerships → referrals to clinics/psychologists/nurseries.

Targeted ads/sponsorships (carefully selected baby products, clinics, etc.).

Key Resources

- Development team (mobile + backend).
- Medical/educational content creators.
- AI/analytics models for growth & sleep analysis.
- Partnerships with clinics, psychologists, nurseries.

Key Activities

- App development & maintenance.
- Content creation (nutrition, parenting scenarios, guides).
- Data analytics (growth/sleep/nutrition).
- Community management.
- Partner onboarding (clinics, specialists).

Key Partners

- Pediatricians, nutritionists, psychologists.
- Clinics & vaccination centers.
- Nurseries & early education centers.
- NGOs focused on child health/mental health.
- Baby product brands (ads/sponsorship).

Cost Structure

- App development & infrastructure (servers, hosting).
- Content production (medical, educational, parenting).
- Expert partnerships (consultants, trainers).
- Marketing & customer acquisition.
- Customer support & community moderation.

Step 3: Unique Selling Proposition (USP)

- Localized Egyptian context: Arabic-first content, local vaccination centers, culturally relevant parenting guidance.
- Interactive parenting scenarios that go beyond static advice.
- Integration of health + nutrition + sleep + behavioral support in one app.
- Special needs support to serve a niche but under-supported segment.

- Partnership potential with local clinics, nurseries, and NGOs.

Step 4: Go-to-Market Strategy

- Early adopters: First-time parents in urban Egypt, especially middle-class families.
- Pilot launch with selected pediatric clinics and nurseries for credibility.
- Marketing through social media platforms (Facebook, Instagram, TikTok) popular among Egyptian parents.
- Referral and word-of-mouth programs to encourage organic growth.
- Freemium approach: Core features free, premium unlocks advanced analytics, sleep solutions, and expert scenarios.
- Gradual scaling to include B2B partnerships with clinics and nurseries.

Step 5: Key Metrics / KPIs

- Daily Active Users (DAU) and Monthly Active Users (MAU).
- User retention rates (weekly and monthly).
- Percentage of free users converting to premium subscriptions.
- Number of consultations booked through the app.
- Engagement levels with trackers (sleep, nutrition, growth).
- Completion rate of parenting scenarios .

Step 6: Detailed Cost Breakdown

1. Development Costs

- Mobile app development (iOS + Android) – 24,000 to 36,000 EGP initial build.
- Backend infrastructure (servers, APIs, cloud hosting) – 35,000 to 65,000 EGP yearly.
- AI/analytics integration (growth, sleep tracking) – 10,000 to 20,000 EGP.
- UI/UX design – 5,000 to 12,000 EGP.

2. Content Creation Costs

- Medical & educational content development – 25,000 to 35,000 EGP yearly.
- Expert validation (pediatricians, psychologists, nutritionists) – 50,000+ EGP yearly.

3. Marketing & Acquisition

- Social media ads & influencer marketing – 25,000 to 50,000 EGP yearly.
- Partnership outreach (clinics, nurseries, NGOs) – 20,000 EGP.
- Referral program incentives – 10,000 to 30,000 EGP yearly.

4. Operations & Support

- Customer support team – 30,000 to 60,000 EGP yearly.
- Community moderators – 30,000 to 50,000 EGP yearly.
- Legal & compliance (medical consultation regulations) – 20,000 to 40,000 EGP.

5. Variable Costs

- Consultation commission fees (per session revenue share with doctors).
- Payment gateway transaction fees (approx. 3–5% per transaction).
- Cloud scaling costs depending on user growth.

References

- [1] Mohammed Bin Rashid School of Government. (2014). *Nazrah 'ala al-i'lam al-ijtima'i fi al-'alam al-'Arabi* A look at social media in the Arab world]. Abu Dhabi: Mohammed Bin Rashid School of Government.
- [2] DeWitt, A., Kientz, J., & Liljenquist, K., "Quality of Mobile Apps for Child Development Support: Search in App Stores and Content Analysis," *JMIR Pediatrics and Parenting*, vol. 5, no. 4, e38793, 2022.
- [3] Anderson, L. M., Shinn, C., Fullilove, M. T., Scrimshaw, S. C., Fielding, J. E., Normand, J., Carande-Kulis, V. G., & Task Force on Community Preventive Services. (2003). The effectiveness of early childhood development programs: A systematic review. *American Journal of Preventive Medicine*, 24(3, Suppl), 32–46.
- [4] Ibrahim, N. M., Ez-Elarab, H. S., Momen, M., Mossad, I. M., & Eletriby, S. S. (2023). A novel wide scale well-baby clinic mobile application: an Egyptian pilot study. *BMC Health Services Research*, 23(687)
- [5] Carlsson, R., Rauti, S., Laato, S., Heino, T., & Leppänen, V. (2023). Privacy in Popular Children's Mobile Applications: A Network Traffic Analysis. In 2023 46th International Convention on Information, Communication and Electronic Technology (MIPRO) (pp.321–326). IEEE. doi:10.23919/MIPRO57284.2023.10159753
- [6] Latharani, T. R., Meghana, G. R., Basavanagowda, T. N., Harshapatel, R. G., Sanjana, A. C., & Sharath, J. M. (2023). *Child Tracking using IoT Device*. *International Journal of Engineering Research & Technology (IJERT)*, 11(05), ICEI 2023.
- [7] Özbek, A. B., & Torppa, M. (2025). Developing an early literacy mobile app: design process and testing the app with children with special needs. *Education and Information Technologies*.
- [8] Grewal, M. K., Rai, K., "Harnessing Mobile Health Technologies to Improve Child Health Outcomes: A Systematic Review of Interventions for Under-5 Year Children in Low- and Middle-Income Countries.," *National Journal of Community Medicine*, vol. 16, no. 4, pp. 413-422, Apr. 2025. DOI: 10.55489/njcm.160420255123.
- [9] David, O. A., Iuga, I. A., Miron, I. S., "Parenting: There is an app for that. A systematic review of parenting interventions apps.," *Children and Youth Services Review*, vol. 156, 2024. DOI: 10.1016/j.chilyouth.2023.107385.
- [10] Indrajaya, M., Chao, W.-H., Semwaiko, G. S., Pusparani, Y., Yang, C.-Y., "Enhancing parent-child interaction through the gamification of parenting: An exploration of digital interventions.," *E-Learning and Digital Media*, vol. 0, no. 0, pp. 1-18, 2024. DOI: 10.1177/20427530241310505.
- [11] Ali, N. M., Ez-Elarab, H. S., Momen, M. A. T., Eletriby, S. S. A., & Shehata, I. M. M. (2024). A novel "Well Baby Clinic Service" mobile application: User requirement assessment and app development. *QJM: An International Journal of Medicine*, 117(Suppl. 1), i240.
- [12] Niklas, F., Annac, E., & Wirth, A. (2020). App-based learning for kindergarten children at home (Learning4Kids): Study protocol for cohort 1 and the kindergarten assessments. *BMC Pediatrics*, 20(554), 1–9. <https://doi.org/10.1186/s12887-020-02447-w>
- [13] Virani, A., Duffett-Leger, L., & Letourneau, N. (2019). Parenting apps review: In search of good quality apps. *mHealth*, 5(44), 1–9. <https://doi.org/10.21037/mhealth.2019.09.09>
- [14] Beňo, L., Kučera, E., & Bašista, M. (2020). Smart sleep monitoring: An integrated application for tracking and analyzing babies' sleep—BabyCare. In *Proceedings of the 2020 IEEE 18th World Symposium on Applied Machine Intelligence and Informatics (SAMII)* (pp. 281–286). IEEE.
- [15] Al-Ahmadi, H. A., "The Effectiveness of Using Smart Device Applications in Developing Some Life Skills for Children with Autism Spectrum Disorder," *Journal of Educational and Psychological Sciences*, vol. 4, no. 20, 2020. DOI: 10.26389/AJSRP.B090520.

[16] Al-Harbi, S. D., “Requirements for Employing Some Artificial Intelligence Applications to Develop Creative Thinking Skills in Kindergarten Children from the Teachers' Point of View,” *The Arab Journal of Specific Education*, no. 19, 2021. DOI: 10.33850/1900-000-019-009.

